

Network Security Fundamentals

Information Security: The practice of protecting digital data both physical with the concept of CIA Triad.

Elements of Information Security:

- Confidentiality
- Integrity
- availability
- authenticity
- non-repudiation

- The security, functionality and usability Triangle

to be able to implement security so that non or technical employee

can use will be following 3 component:

- Functionality: Feature provided by the system
- Usability: GUI for ease of usage
- Security: restriction with accessing and depending of system
- + NIST CSF consists of key component such core, tiers, profiles and implementation guideline and for the function which the goals are Identify, Protect, Detect, Respond and Recover
- + Impact of Security attacks are Financial losses, loss of confidentiality and integrity, damaged customer relationship, loss of business reputation legal and compliance issues and operational impacts.

+ Information Assurance: the practice of implementation and managing risk to defend and give reliability through lifecycle which resulted in increase profits, improved productivity, enhanced compliance and client confidence both trust.

+ Type of Network defense approach which are to preventive, Reactive, Retrospective and proactive.

+ Continual/Adaptive Security Strategy which include protect, detect, respond, predict.

+ Technical Security Controls includes system access control, network access control, authentication, authorization, encryption and protocols, network security device, auditing.

+ Network defense element:

- Technology
- Operations which consists of creating and enforcing policy, create and enforce standard network operating procedures, planning business and continuity, configure control management, create and IR.

- People which consists of

- Network defense team
 - network administrator
 - network security admin
 - network security engineer
 - security architecture
 - security analyst
 - network technician
 - end user
- Leadership

• Incident handling and response team

- info sec officer (ISO)
- incident manager (IM)
- incident coordinator
- forensics investigator
- threats researcher
- system admin
- network admin
- internal auditor
- financial auditor
- human resource
- public relation

• Computer forensics investigation team

- photographer
- IR
- incident responder
- evidence examiner/investigator
- evidence documenter
- evidence manager
- expert witness
- attorney